

High Stock Portfolio Returns? Easy

by: Guy R. Fleury

High Stock Portfolio Returns? Easy will demonstrate that among your long-term investment choices, at least one strategy could propel you to unbelievable heights. And with little to do to get there.

The strategy is free and will be detailed at length. Your skills, your means, and the choices you are to make, if any, will determine the outcome. It is all doable. You decide how far you want to go.

In any stock portfolio, you enter trades and get out of them sometime later. It is almost the definition of a stock trade. You have an entry and an exit price. What you are interested in is the outcome. Will you make a profit or not on that trade? And that alone raises a lot of questions.

If we have equations that can fit the outcome of our trading strategies, does it not stand to reason that whatever or however those strategies trade, we should end up satisfying those equations?

We can classify the outcome of any trade as a gain or a loss of $\pm x_i$, for $i = 1$ to N . Any such trade could be made at any time for any amount for any reason for any duration. Both you or your program can make those entry and exit decisions. Both operations will take a fraction of a second. Your computer can do the same job in microseconds. Your competition can do it in nanoseconds using better hardware and more sophisticated trading programs.

It is not the execution that makes you money. To have a profit or loss, you need a price difference. It is the waiting time between entry and exit that makes that difference. That time interval could be of any duration you want, from a few seconds to decades.

The profit or loss does not identify itself beforehand. It is just a money amount. In the sense that $\pm x_i$ could come from any trade i , and as was said, from trade $i = 1$ to N , which could be in the thousands over the years.

How many trades could you make based on available capital, the number of potential and feasible trades, and time constraints? If you intend to trade for only one year, I suggest you forget about it unless you feel very lucky or know something of value market-wise.

My previous article titled [Your Stock Trading Portfolio Destroyer](#) made the point clear that the random side of a stochastic equation, without its drift component, is

bound to destroy your portfolio trading short-term over the long-term and that over the short-term, it remains a crapshoot. You can't predict that good that the next day is an up day or a down day. You always end up with a hard maybe or some platitude like: "It will rise tomorrow if it does not fall".

Have A Plan

One thing you could do is have a plan to determine the purpose of your investment fund and its duration. It immediately sets three prerequisites: time, available capital, and profitable trading methods.

We could express it all as $F(t) = F_0 \cdot (1 + \bar{r})^t$, with F_0 your initial capital, and $\bar{r}$ your average rate of return over the period t . It gives you a starting value and its outcome. What you need is a value for $\bar{r}$. The equation also stands for a fully invested portfolio over the interval t .

You could use the same expression for a stock portfolio; you sum all the trades up $F(t) = \sum_{i=1}^N f_{i-1} \cdot r_i$ where the sum of all profits and losses will be accounted for.

You have for any position: f_{i-1} , the value of your portfolio from the previous period. You need trading methods to give you a positive return $+r_i$. As your portfolio grows, so will your bet size. If you have drawdowns, your bet size is automatically reduced. The intent is to reduce the position size on drawdowns and increase it on the long side as the portfolio increases in value.

The constraints are set. You have limited funds, limited time, limited trade availability, and limited methods of profit extraction. Within that time limit, you have to be able to do the job, which is not the same as you have or hope to do the job.

We can have a hard time predicting the immediate future, but not so much the general market outcome over the long term. You should expect to get the market's long-term average return by holding a market proxy over many years. Holding SPY for 30 years will get you SPY's long-term average return $\bar{r}_m$, but nothing more.

It's not a coincidence, but that is what the average money manager can achieve in the long term. Up to 75% do not even reach that goal, which is almost given away for doing nothing but sitting on their hands. A lot of closet indexers do the same.

We could sum all the trades you might do in the future using the payoff matrix notation: $F(t) = F_0 + \sum_1^N (\mathbf{H} \cdot \Delta \mathbf{P})$ where $\mathbf{H}$ is the stock inventory holdings and $\mathbf{P}$ the price matrix for each stock in your portfolio, while $\Delta \mathbf{P}$ is the price difference matrix from period to period. The price difference matrix includes the entry, exit, and prices in between. You could have thousands of trades in that payoff matrix. The individual profit or loss could be expressed as $\pm x_{(i+\tau),j} = h_{i,j} \cdot \Delta p_{i,j}$.

At all times, you know the value of your stock portfolio by either summing horizontally any row to get the portfolio value or vertically to get the contribution of each stock to the total outcome.

All trading strategies will end with: $F(t) = F_0 + \sum_1^N (\mathbf{H} \cdot \Delta \mathbf{P})$. You cannot do much about $\mathbf{P}$, the price matrix. $\Delta \mathbf{P}$ gives the amount of profit or loss on a trade. You also have the positions' return matrix as $\frac{\Delta \mathbf{P}}{\mathbf{P}} = \mathbf{R}$.

For any element of the payoff matrix, we have: $\frac{\Delta_i p_{i,j}}{p_{i,j}} = \pm r_{i,j}$. So, your profit or loss is: $q_{i,j} \cdot \Delta_i p_{i,j} = q_{i,j} \cdot p_{i,j} \cdot r_{i,j} = x_{i,j}$. From there, $\sum_{i,j} x_{i,j} = X$, the total profit or loss generated by that trading strategy or method over the portfolio's entire stay in the market, no matter how many positions were taken and no matter how they were scattered over the timeline. The portfolio outcome will be $F(t) = F_0 + X$.

Will X be positive or negative? That is entirely up to you.

Nothing says which stock was taken or for what reason it was selected. Nor did we mention a price other than an initial entry price, whatever it might have been. You could have chosen any stock you wanted for any reason whatsoever. For your trade, it would only be an entry price $p_{i,j}$ on a particular stock j at a specific time before T , the portfolio's termination time.

We know that $\frac{\sum_1^N (\mathbf{H} \cdot \Delta \mathbf{P})}{N} = \bar{x}$, the average profit per trade. We also know the average growth rate:

$$\left[\frac{F_0 + \sum_1^N (\mathbf{H} \cdot \Delta \mathbf{P})}{F_0} \right]^{\frac{1}{t}} - 1 = \bar{g} \quad (1)$$

What must we do in our trading strategy to get an outcome near $\bar{x}$ or $\bar{g}$?
Is anything in there under our control?

We only have the payoff matrix to contend with, and that is simply the trading strategy or trading methods we want to implement. The payoff matrix is a major contender in these equations. It gives the cumulated profits and losses of all the trades taken over the entire trading interval, whether those positions were closed or still opened. The outcome even includes partial trades.

Could we not walk backward from the wanted results and design our trading strategy to perform as we would want it to? Do we have clues that could help in this determination?

A Strategy Example

With the **One Percent Per Week** TQQQ trading strategy, the request or objective was to reach 1% return per week.

You could determine before you started how much, on average, it should produce per year: $F(t) = F_0 \cdot (1 + 0.01)^{52} = 1.6777 \cdot F_0$. Should you reach the objective, you would be making 67.77% compounded per year.

If we compare it to average market performance, it could be considered astronomical but not unprecedented. It is at the same level as Renaissance's Medallion Fund over the last 30 years before fees and expenses (4/44). The general market can provide, on average, over the long term about 10%.

If you want more, you have to bring some trading skills to the game.

Should you put less on the table, you should gain less: $0.5 \cdot F_0 \cdot (1 + 0.01)^{52} = 1.3388 \cdot F_0$. Still, that is a 33.88% compounded return. You would only get half a fully invested scenario. Therefore, to benefit from your investment portfolio, you should aim for full exposure; otherwise, it will reduce your strategy's potential.

From the start, we also know the number of trades to be executed in the **One Percent Per Week** trading strategy.

Its trading rules are simple:

- buy on Monday at the open,
- sell if profit targets of 7% or 8% are hit before Friday's close,
- sell at Friday's close if the profit targets were not hit.

It makes one trade per week. Over 15 years, you should expect $15 \cdot 52 = 780$ trades.

The TQQQ strategy has a time-limit stop-loss built-in. All trades are limited to 5 trading days and no longer. Every Monday, you enter a trade, and at the latest, the position is liquidated at Friday's close.

That does not stop the strategy from having drawdowns, but they might be less pronounced due to the lower exposure. However, the strategy is in a 3x-leveraged scenario, and its average standard deviation will be 3x larger than that of QQQ, as should be expected.

We could evaluate the outcome of the **One Percent Per Week** strategy over those 15 years: $F_0 \cdot (1.0 + 0.01)^{780} = 2,347.85 \cdot F_0$ which for a \$100k starting portfolio would give \$234,785,650. Note that the strategy is presented as fully invested, which is not the case based on simulation results.

You have not done anything yet, and you already know the expected outcome of your trading strategy.

Your 15-year objective is to reach \$100+ million.

That is based on your strategy design, meaning that your trading procedures will have to generate, on average, that 1% per week.

It is not requested that you make 1% per week every week but to make, on average, the equivalent of 1% per week.

In the last presented simulation, the average percent profit per trade was 1.03%. So, that 1% per week objective stood for 14.4 years or for 753 weeks.

You are left with finding the tools to do the job. And that is relatively simple, too.

You already have a solution in the free **One Percent Per Week** trading strategy.

But before you use that published strategy, put in the modifications suggested in **Parts I** and **II** of my article series. You will need them to increase performance to the desired level. Those code modifications are easy to do, as described in those two articles.

Better yet, find ways to improve on that strategy. Find ways to add some downside protection since you will come to need it. I do not know when, but there will come a point where you will realize that you needed it. By then, you might find you should have done so before the downfall happened. Determining those protective measures is up to you, but they should be addressed.

You can improve that strategy by adding downside protection and improving the average return per trade.

Say you find ways to increase the average percent profit per trade to 1.25% per week.

A quart of a point more appears insignificant, but if we apply it to the above equation, and being fully invested over those 15 years, we would get:

$$\$100,000 \cdot (1.0 + 0.01 + 0.0025)^{780} = \$1,614,822,796$$

That quart-point quest just became significant.

The strategy improvement could be considered minimal. Adding a quarter point in the above scenario increased the outcome 6.87 times, which is money too. This quarter point is on the average return. It's like having SPY with its long-term ~10% average and seeking to increase it to 12% or 15%.

Is it worth it to find ways to get that added quart point?

How about adding another quart point:

$$\$100,000 \cdot (1.0 + 0.01 + 0.0025 + 0.0025)^{780} = \$11,053,834,409$$

The quest is not about finding extraordinary trading methods; it is about making money, and we should look at methods and procedures that are susceptible to making it so.

You are not looking for big moves but for ways to increase the average return per period. You want to add more alpha to your portfolio: $F(t) \equiv F_0 \cdot (1 + \bar{r}_m + \alpha_1 + \alpha_2)^t$. You will find your added alpha in your trading procedures.

Since you can do a simulation to see what your trading methods would have done, you could extract those average performance metrics and make future estimates based on them, especially if the number of trades is statistically significant, as in this scenario.

The math of your trading strategy tells you what is important, including how and where to improve it.

For example, in the **One Percent Per Week** strategy, the $\pm\%$ average variation per trade was expected to be about $\pm 2.56\%$. Weekly variations were larger than this, but, as stated, market exposure was lower. We can make that claim since we are looking at a quasi-normal distribution of returns, and the average percent per trade should be about the same on both sides of the bell-shaped curve.

The numbers from the last simulation with its 50% exposure were:

$$0.50 \cdot \$100,000 \cdot (1 + 0.0453)^{386} \cdot (1 - 0.0265)^{367} = \$70,028,150$$

After its 753 trades, the strategy simulation generated \$70,799,226 in profits. It is close enough if we consider the rounding of the return rates. Over the simulation period, that is an average 57.61% CAGR.

We should consider the number of trades sufficient to be representative of the strategy's behavior. It had a 51.26% hit rate, which is not that far from its 50/50 initial expectation. It was as if the Monday trades were taken on a random basis. Refer to Figure #2 in [Welcome To YOUR Stupendous Retirement Fund](#) for other details and simulation results.

Taking a long position every Monday is the same as playing heads on every flip of a fair coin. And therefore, we should catch whatever underlying sequence of heads the time series has to offer.

We already know that QQQ, SPY, or DIA are slightly biased to the upside with up days at about 51% to 52%. Therefore, flipping that coin heads every Monday is not gambling. It is playing the numbers. All your bets were in the same direction. Whether you made a profit or not on those trades was another matter.

Improving The Strategy

You take the **One Percent Per Week** strategy as presented in the above-cited article and improve the average percent return per winning trade by 10%. Using the above strategy equation, you would get:

$$0.50 \cdot \$100,000 \cdot (1 + 0.0453 \cdot 1.10)^{386} \cdot (1 - 0.0265)^{367} = \$371,699,271$$

Figures #1 and #2 in [Welcome To YOUR Stupendous Retirement Fund](#) showed that strategy had profits of \$70,799,226 over those 14.4 years. Improving the average percent per winning trade by 10% would have made the portfolio grow to \$371,699,271. You did not ask for a major increase in performance, and yet the strategy would dramatically increase its outcome to a 76.98% CAGR.

Should we make that extra effort to get that added 10% and go from 0.0453 to 0.0498 on the average percent per winning trade?

It could be considered a minimal request. Adding another alpha source to the equation would do it. $F(t) \equiv F_0 \cdot (1 + \bar{r}_m + \alpha_1 + \alpha_2 + \alpha_3 + \dots)^t$.

If you reduced the average percent loss per losing trade, the strategy would also benefit from your effort:

$$0.50 \cdot \$100,000 \cdot (1 + 0.0453 \cdot 1.10)^{386} \cdot (1 - 0.0265 \cdot 0.90)^{367} = \$1,008,028,760$$

This move reduces the average percent lost per losing trade from 0.02650 to 0.02385. Again, this is not a major request but will undoubtedly show its long-term impact. It would have increased the overall CAGR to 89.68%.

All our concentration should be on the average percent win on winning trades and the average percent loss on losing trades. And since we are dealing with long-term averages, these will settle long-term to those designed within the trading procedures over the first 50 to 100 trades.

After some 753 trades, those averages will have settled down to their long-term averages.

The average return per trade difference would tend to be close to $\bar{r}$ after so many trades: $\frac{r_i - \bar{r}}{\bar{r}} \cdot \frac{1}{753} \rightarrow \bar{r}$. We would have 1/753 of the net change applied to $\bar{r}$. That is not something that will add much pressure to the long-term average.

We are back with the product equation: $F(t) = F_0 \cdot \prod_1^N (1 \pm r_i)$ where every r_i counts. We could also reformulate it as: $F(t) = F_0 \cdot (1 + \bar{r}_+)^{N-\lambda} \cdot (1 - \bar{r}_-)^{\lambda}$ where you have $N - \lambda$ winning trades with an average percent per trade of $\bar{r}_+$ and λ losing trades with an average percent loss per trade $\bar{r}_-$. The same formula was used to express the simulation result.

But there is still more to it.

You could increase the exposure rate by 50%, making it 75%. You will not reach 100% exposure due to the execution of the profit targets before Friday's close. You could change the strategy's trading rules to gain more exposure.

Still, a 75% average exposure rate would be great. It would raise the CAGR to 95.10%.

$$0.50 \cdot 1.50 \cdot \$100,000 \cdot (1 + 0.0453 \cdot 1.10)^{386} \cdot (1 - 0.0265 \cdot 0.90)^{367} = \$1,512,043,139$$

You could add more time.

You already know the program makes one trade per week. If you extend it five years, this would add $5 \cdot 52 = 260$ trades.

Since the strategy has had 753 trades to settle its long-term averages, we could use the same win rate as given by the last simulation, which is about the same as the long-term market average (a 200+ year average). This would not be a stretch from historical data; it would be the long-term average. It would add 133 trades to the winning trades and 127 to the losing side.

$$0.50 \cdot 1.50 \cdot \$100,000 \cdot (1 + 0.0453 \cdot 1.10)^{(386+133)} \cdot (1 - 0.0265 \cdot 0.90)^{(367+127)} = \$45,393,256,260$$

After some 19.44 years, you would be expected to reach the 45 billion mark. It would amount to having had a 95.70% CAGR over those 19.44 years.

You could also use leverage.

That is the same as increasing the exposure rate. Doubling the exposure would give 1.50 as effective leverage: $0.50 \cdot 1.50 \cdot 2.0 = 1.5$. And the long-term impact would also be visible. It would propel the strategy's CAGR to 102.82% over those 19.44 years.

$$0.50 \cdot 1.50 \cdot 2.0 \cdot \$100,000 \cdot (1 + 0.0453 \cdot 1.10)^{(386+133)} \cdot (1 - 0.0265 \cdot 0.90)^{(367+127)} = \$90,786,512,520$$

All the above is based on the same TQQQ strategy with minor modifications to its code to increase the trading interval, market exposure (including leverage), and the spread between the winning and losing percent per trade.

The above is feasible because the short-term trade interval is limited to one week.

The volatility at that scale is sufficient to drive this TQQQ strategy.

The strategy tries to take advantage of the short-term volatility. TQQQ is three times more volatile than QQQ by definition. And since QQQ has a weekly volatility of around 5%, we have to expect that TQQQ's volatility will be around 15% per week.

It is the reason why the profit targets were set at 7% and 8%. This way, they could often be reached and, at times, even exceeded. The TQQQ strategy is simply playing on this ETF's volatility.

It is secured by the underlying NDX index (QQQ being its tradable version).

Technically, you are playing QQQ on steroids (3x-leveraged) without paying for that leverage. Only the added leverage would generate leveraging fees. Overall, even if you leveraged TQQQ by a factor of 2.0, you would only pay for the excess 2x you applied to the strategy. The leveraging fees would be less due to the 75% average exposure.

You effectively leveraged QQQ 6x times, most of the time. No wonder you can increase this strategy's performance. It also gives you even more reasons to put in some downside protection. You will need it.

The weekly 7% and 8% profit targets limit this strategy to high-volatility stocks. It would not produce as much if used on QQQ, for instance. You would lack the leverage and volatility. You could make some adjustments. But you could only leverage QQQ up to 2x, not 6x.

The TQQQ strategy, as presented in the **One Percent Per Week**, is well suited to make it an outstanding trading strategy, having the reassurance that it is based on the top 100 highest-valued stocks in QQQ.

Like in many of my articles, it all comes back to you. What will you do about it? Which level in the above scenarios will you be going for, if any? And how about going even higher? For instance, adding options to the above scenarios could increase overall CAGR.

Note that simply adding a zero to the initial stake would multiply all those results by a factor of 10. That, too, is up to you.

Related Papers and Articles:

[Your Stock Trading Portfolio Destroyer](#)

[You Will Earn Every Penny You Make](#)

[Stock Trading Strategy Alpha Generation](#)

There Is Always A Better Retirement Fund: [Part I](#), and [Part II](#)

[Welcome To YOUR Stupendous Retirement Fund](#)

The One Percent a Week Stock Trading Program: [Part VII](#), and [Part VIII](#)

The One Percent a Week Stock Trading Program: [Part V](#), and [Part VI](#)

The One Percent a Week Stock Trading Program: [Part III](#), and [Part IV](#)

The One Percent a Week Stock Trading Program: [Part I](#), and [Part II](#)

The Long-Term Stock Trading Problem: [Part I](#), and [Part II](#)

[The MoonPhaser Stock Trading Program](#)

[Anticipating A Stock Portfolio's Long-Term Outcome](#)

[The Big Open Project](#)

[Sitting On Your Bunnies Might Be Your Best Investment Yet](#)

[Self-Managed Retirement Funds](#)

[Make Yourself A Glorious Retirement Fund](#)

[The Age Of The Individual Investor](#)

[QQQ To The Rescue](#)

[Take the Money and Keep it – II](#)

[Use QQQ - Make the Money and Keep IT](#)